

Safe Band: IoT-Based Smart Wearable for Emergency Health Alerts and Monitoring in Sri Lanka

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Project Proposal Report

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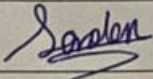
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Cardiovascular diseases, respiratory failures, and sudden cardiac events represent leading causes of preventable death globally, with Sri Lanka reporting over 30,000 cardiac related deaths annually. A critical challenge in emergency healthcare is the delay between the onset of a medical emergency and the delivery of appropriate care, largely due to the absence of real time, automated physiological monitoring systems for at risk patients. Existing consumer wearable devices lack the clinical intelligence to detect critical health events autonomously and alert emergency services without human intervention.

This proposal presents the ML Driven Emergency Detection System, a core component of the broader Safe Band research initiative. This component leverages a smartwatch to continuously collect vital physiological parameters including heart rate (HR), blood pressure (BP), blood oxygen saturation (SpO2), and electrocardiogram (ECG) signals from patients. The collected data is processed through a multi class Machine Learning classification pipeline deployed on an Android mobile application, which is designed to detect potential critical health conditions such as cardiac arrest, hypertensive crises, hypoxia, and atrial fibrillation in near real time.

The proposed system integrates with Firebase Realtime Database for cloud synchronization, enabling timely alerts to both the Android application and the ML based ambulance dispatch component upon detecting a potential critical event. Algorithms under evaluation include Long Short Term Memory (LSTM) networks, Random Forest, and ensemble methods trained on annotated clinical datasets. The system aims to achieve reliable detection performance with low latency and is intended to be evaluated against clinically relevant benchmarks within the Sri Lankan healthcare context.

The system directly supports UN Sustainable Development Goal 3 (Good Health and Wellbeing) and addresses the gap between consumer health wearables and clinical grade emergency detection in resource limited settings. It has potential for future commercialization as a subscription based telehealth monitoring service.

Keywords: Machine Learning, Emergency Detection, IoT Wearable, ECG Analysis, Heart Rate Monitoring, SpO2, Firebase, Smartwatch, Sri Lanka Healthcare, Real Time Health Alert.

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TABLE OF CONTENT (Auto Generated)

Contents

DECLARATION	2
ABSTRACT	3
ACKNOWLEDGEMENT	4
TABLE OF CONTENT (Auto Generated)	5
LIST OF TABLES	6
LIST OF FIGURES	7
LIST OF ABBREVIATIONS	8
LIST OF APPENDICES	9
1. INTRODUCTION	10
1.1 Background and Context.....	10
1.2 Literature Review	11
1.3 Research Gap	13
2. OBJECTIVE	15
2.1 Main Objective.....	15
2.2 Specific Objectives	15
3. METHODOLOGY	16
4. HIGH-LEVEL SYSTEM ARCHITECTURE	18
5. USER REQUIREMENTS.....	22
6. COMMERCIALIZATION PLAN	24
7. BUDGET AND JUSTIFICATION.....	25
8. WORK BREAKDOWN STRUCTURE (WBS).....	26
9. GANTT CHART.....	28
10. REFERENCES LIST.....	29
11. APPENDICES.....	31

LIST OF TABLES

Table 1: List of abbreviations	8
Table 2: Tools and Platforms.....	17
Table 3: System Components.....	18
Table 4: Budget and justification.....	26
Table 5: Work Breakdown Structure (WBS).....	27

LIST OF FIGURES

Figure 1: System diagram of components.....	18
Figure 2: Preliminary UI.....	31

LIST OF ABBREVIATIONS

API	Application Programming Interface
BP	Blood Pressure
ECG	Electrocardiogram
Firebase	Google Firebase Realtime Database & Cloud Services
HR	Heart Rate
IoT	Internet of Things
LSTM	Long Short-Term Memory (Neural Network)
ML	Machine Learning
RF	Random Forest
SDG	Sustainable Development Goal
SpO2	Peripheral Oxygen Saturation
UI	User Interface
WearOS	Wear OS (Android-based wearable operating system)

Table 1: List of abbreviations

LIST OF APPENDICES

Preliminary UI mockups.....	31
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1. INTRODUCTION

1.1 Background and Context

Cardiovascular disease (CVD) remains one of the leading causes of mortality globally. In Sri Lanka, non communicable diseases, particularly cardiac and respiratory conditions, account for a significant proportion of total deaths, with the Ministry of Health (2022) reporting over 30,000 cardiac related fatalities annually. A considerable number of these deaths are associated not only with the severity of underlying conditions but also with delays in emergency response and the lack of continuous physiological monitoring for at risk individuals.

Patients with chronic conditions such as hypertension, heart failure, cardiac arrhythmia, and chronic obstructive pulmonary disease (COPD) require continuous monitoring of key physiological parameters including heart rate, blood pressure, blood oxygen saturation (SpO₂), and electrocardiographic (ECG) activity to enable early detection of potential deterioration. However, in the current Sri Lankan healthcare context, such monitoring is largely limited to periodic clinical visits or inpatient care settings. As a result, patients remain vulnerable during extended periods without clinical supervision, increasing the risk of undetected acute events.

Existing remote patient monitoring solutions range from hospital based telemetry systems to consumer grade wearable devices. Hospital telemetry systems provide clinically reliable data but are restricted to inpatient environments and are not accessible for continuous community based monitoring. Conversely, mobile health (mHealth) applications and wearable devices extend monitoring beyond clinical settings but are primarily limited to data collection and basic visualization. These systems generally lack the capability to autonomously interpret multi parameter physiological data, identify emergency conditions in real time, or generate context aware alerts. Furthermore, limited integration with emergency response systems, such as ambulance dispatch and hospital workflows, reduces their practical effectiveness, particularly in resource constrained environments such as Sri Lanka.

Recent advancements in wearable technologies, particularly smartwatches equipped with biosensors, present an opportunity to bridge this gap. These devices are capable of continuously capturing multiple physiological signals in a non invasive manner within everyday environments. However, despite their sensing capabilities, most existing wearable devices lack embedded clinical intelligence required for real time analysis, emergency classification, and automated alert generation. This disconnect between data acquisition and actionable decision making forms the central motivation for this research.

In response to these challenges, the Safe Band project is proposed as a multi component research initiative aimed at developing an integrated IoT based emergency health

monitoring ecosystem. The system comprises four key components: (1) an Intelligent AI-Based Hospital Ward Monitoring System, (2) a Health Progress Prediction Module, (3) an ML Driven Emergency Detection System, and (4) an ML-Based Smart Ambulance Dispatch System. This research focuses specifically on Component 3, the ML-Driven Emergency Detection System, which serves as the core analytical layer responsible for processing continuous physiological data and identifying potential emergency health states. The component is designed to apply machine learning techniques to detect conditions such as cardiac abnormalities, arrhythmias, and hypoxic events, and to generate timely alerts for downstream systems including emergency response services.

This research aligns with the United Nations Sustainable Development Goal 3 (Good Health and Well-being), particularly Target 3.4, which emphasizes reducing premature mortality from non-communicable diseases, and Target 3.8, which promotes access to quality healthcare services. By enabling continuous monitoring and automated emergency detection, the proposed system aims to contribute toward improving healthcare accessibility and reducing preventable mortality in resource-limited settings.

1.2 Literature Review

A substantial body of literature has examined the application of machine learning to physiological signal analysis and emergency health detection. The following review synthesizes key findings across four thematic areas: (i) ML-based ECG and cardiac arrhythmia detection, (ii) wearable SpO₂ and blood pressure monitoring, (iii) multi-parameter fusion and IoT health monitoring frameworks, and (iv) existing commercial products and their limitations.

ML-Based ECG and Cardiac Arrhythmia Detection

The automated interpretation of electrocardiographic (ECG) signals through deep learning has been one of the most extensively studied areas in clinical AI. Rajpurkar et al. (2017) demonstrated that a deep convolutional neural network could detect 12 types of cardiac arrhythmia from single-lead ECG data with accuracy at or surpassing that of board-certified cardiologists, achieving an F1 score that exceeded the average cardiologist performance benchmark [1]. This landmark study established the foundational feasibility of automated ECG interpretation and remains a primary reference for AI-assisted cardiac monitoring systems. Building upon this foundation, Hannun et al. (2019) extended arrhythmia classification to a 14-class problem using a 34-layer residual neural network trained on 91,232 single-lead ECG recordings, achieving a mean area under the receiver operating characteristic curve (AUC) of 0.97 [2]. The scalability and accuracy of this approach demonstrated the viability of deep residual architectures for real-time ECG classification in wearable contexts, directly informing the model selection rationale in the present research. At the clinical application level, the Apple Heart Study (Perez et al., 2019) provided large-scale population evidence by monitoring over 419,000 participants using a consumer smartwatch. The study demonstrated irregular pulse notifications consistent with atrial fibrillation (AF) in 0.52% of participants, with a positive predictive value of 84% when confirmed by ECG patch [4]. Topol (2019) further reviewed the broader landscape of ML applications in medicine, highlighting that wearable sensor-based AF

detection algorithms had achieved sensitivity exceeding 98% across multiple validation cohorts [3]. Collectively, these findings confirm that real-time cardiac event detection from wearable ECG is clinically achievable at scale.

Wearable SpO2 and Blood Pressure Monitoring

Beyond cardiac rhythm, continuous monitoring of peripheral oxygen saturation (SpO2) and blood pressure constitutes an essential capability for detecting respiratory deterioration and hypertensive emergencies. Luks and Swenson (2020) conducted a systematic review of the clinical utility of consumer-grade pulse oximetry, finding that wearable SpO2 sensors had achieved accuracy within $\pm 2\%$ of laboratory co-oximetry in controlled validation studies [5]. This level of accuracy is clinically sufficient for generating early warning alerts for conditions such as hypoxaemia associated with COPD exacerbations or acute respiratory failure. In the domain of cuffless blood pressure estimation, Kario et al. (2021) evaluated photoplethysmography (PPG)-based wearable blood pressure monitors and demonstrated that appropriately calibrated algorithms could achieve a mean absolute error (MAE) below 5 mmHg the threshold mandated by ISO 81060-2 clinical accuracy standards [6]. This finding is directly relevant to the Safe Band architecture, which incorporates PPG-derived blood pressure estimation from the Smart Watch 8 as one of its four primary physiological input parameters.

Multi-Parameter Fusion and IoT Health Monitoring Frameworks

Single-parameter monitoring systems are inherently limited in their capacity to detect complex, multifactorial emergency health states. Clifford et al. (2016) demonstrated that combining heart rate (HR), SpO2, and respiratory rate in an ensemble model for early warning score computation outperformed single-parameter models by 23% in specificity [7]. This finding provides a direct empirical basis for the multi-parameter classification architecture proposed in this research, in which HR, SpO2, blood pressure, and ECG signals are fused as a composite input feature vector to a machine learning classifier. In the IoT infrastructure domain, Islam et al. (2020) proposed a cloud-based health monitoring framework integrating wearable sensors with Firebase Realtime Database, achieving end-to-end alert latency of under 8 seconds from sensor acquisition to emergency notification [8]. This architecture closely parallels the Firebase-centric cloud integration approach adopted in the Safe Band ecosystem, providing validated evidence that sub-10-second alert delivery is achievable in cloud-based wearable health monitoring systems. With respect to temporal modelling of physiological time series, Lipton et al. (2016) demonstrated that long short-term memory (LSTM) recurrent neural networks applied to multivariate clinical time series outperformed traditional classifiers including logistic regression and random forests on intensive care unit (ICU) mortality prediction tasks [9]. The ability of LSTM networks to capture sequential dependencies in continuous physiological data streams establishes the theoretical basis for their application to wearable sensor data in the present research.

Existing Commercial Products and Their Limitations

The current state of the art in consumer health wearables is represented by devices such as the Apple Watch Series 9 (ECG and AF detection), Fitbit Sense 2 (stress monitoring and SpO₂), and Samsung Galaxy Watch 6 (PPG-based blood pressure estimation and ECG). While these platforms demonstrate the technical feasibility of wearable physiological sensing, Bayoumy et al. (2021) conducted a systematic evaluation of consumer health wearables and identified several critical limitations that preclude their use as clinical emergency detection systems [10]:

- i. These devices are designed and validated for wellness tracking and fitness monitoring rather than clinical emergency detection and consequently do not implement emergency classification logic.
- ii. None of the reviewed devices integrate with hospital information systems (HIS), emergency dispatch platforms, or real-time clinical alert workflows.
- iii. Existing devices do not perform multi-parameter emergency state classification they report individual sensor readings without synthesizing a composite emergency risk score.
- iv. No current commercial wearable has been validated in Sri Lankan or broader developing-country clinical contexts, where population-specific physiological baselines and environmental factors may differ from those in which the devices were designed and tested.

1.3 Research Gap

The literature review indicates that although substantial progress has been achieved in machine learning-based cardiac arrhythmia detection and physiological monitoring systems, several critical research gaps remain. Existing studies primarily focus on single-parameter analysis, such as heart rate or ECG, without effectively integrating multiple physiological parameters including heart rate (HR), blood pressure (BP), oxygen saturation (SpO₂), and electrocardiogram (ECG) into a unified machine learning pipeline. Moreover, current commercial smartwatch solutions are limited in functionality, often relying on single-parameter detection (e.g., atrial fibrillation via ECG) and failing to identify complex or composite emergency conditions such as simultaneous hypoxia and hypertensive crises.

In addition, most existing systems are developed and validated in non-local contexts, with no evidence of machine learning-driven emergency detection systems specifically tailored to the physiological characteristics and healthcare infrastructure of Sri Lanka. Current IoT-based health monitoring frameworks also lack integration with real-time emergency response mechanisms, such as hospital alert systems and ambulance dispatch services, resulting in delayed clinical intervention. Furthermore, many consumer wearable devices depend on static, rule-based threshold systems rather than adaptive machine learning models, which limit their ability to detect early-stage physiological deterioration with high accuracy.

Therefore, this research addresses these limitations by proposing a novel, multi-parameter machine learning-based emergency detection system that utilizes smartwatch-derived data within an Android-based ecosystem. The system incorporates multi-modal sensor fusion and

sequential analysis using advanced models such as Long Short-Term Memory (LSTM) networks to enable real-time detection of critical health conditions. Additionally, it integrates with a cloud-based infrastructure to support rapid alerting healthcare providers and emergency services, ensuring timely intervention. The novelty of this approach lies in its combination of multi-parameter analysis, real-time processing, and end-to-end integration with emergency response systems, specifically adapted to the Sri Lankan healthcare context.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a machine learning-based emergency detection system that utilizes continuous physiological data (heart rate, blood pressure, blood oxygen saturation (SpO₂), and ECG) collected from wearable devices to identify potential critical health events and generate timely alerts within the Safe Band IoT-based health monitoring ecosystem in the Sri Lankan context.

2.2 Specific Objectives

- i. To develop a wearable-integrated data acquisition framework that enables continuous collection of physiological parameters including heart rate, blood pressure, SpO₂, and ECG signals.
- ii. To design and implement an effective data preprocessing and feature extraction approach suitable for multi-parameter physiological time-series data.
- iii. To develop and compare multiple machine learning models, including deep learning and traditional approaches, for the classification of potential emergency health conditions such as cardiac abnormalities, hypertensive events, and hypoxic conditions.
- iv. To implement the selected machine learning model within a mobile-based application to support real-time or near real-time inference.
- v. To design and integrate a cloud-based alerting mechanism that enables communication between the detection system and relevant healthcare or emergency response components.
- vi. To evaluate the overall system performance in terms of detection capability, response time, and usability under controlled experimental conditions.

3. METHODOLOGY

This research adopts a Design Science Research (DSR) methodology, which is well suited for the development and evaluation of IT artefacts. The study follows six key phases: Problem Identification, Solution Design, Artefact Development, Demonstration, Evaluation, and Communication.

Research Approach

The Design Science Research approach was selected because the primary deliverable is a functional software and ML artefact (the emergency detection system) rather than a purely theoretical contribution. This approach demands iterative design-build-test cycles, making it well-suited to ML model development where algorithm selection, feature engineering, and threshold calibration require empirical refinement.

Data Collection

Physiological data for this research will be obtained from two primary sources. Firstly, publicly available annotated clinical datasets will be used for initial model training and validation. These datasets include established repositories containing electrocardiogram (ECG) signals and vital parameters such as heart rate, blood pressure, and blood oxygen saturation.

Secondly, wearable device-based data collection will be conducted using volunteer participants to support system testing and contextual validation. This process will be conducted under appropriate ethical approval.

Ethical considerations will include obtaining informed consent from all participants, ensuring anonymization of collected data, and maintaining secure data storage using controlled-access cloud infrastructure. No personally identifiable information will be used for analysis or publication purposes. Additionally, emergency detection scenarios will be evaluated using simulated or pre-recorded physiological data to ensure participant safety.

Technical Methods and Algorithms

The machine learning pipeline will be developed using standard data science frameworks and later adapted for deployment within a mobile environment. Multiple machine learning approaches will be explored, including deep learning and traditional classification techniques.

Long Short-Term Memory (LSTM) networks will be considered for modeling temporal dependencies in physiological time-series data. In addition, traditional machine learning algorithms such as Random Forest will be used as baseline models for comparison. Ensemble techniques may also be explored to improve overall classification robustness.

Data preprocessing and feature extraction techniques will be applied to transform raw physiological signals into meaningful input representations for model training. The system is designed to classify multiple health states, including normal and potential emergency conditions related to cardiac and respiratory abnormalities.

Tools and Platforms

Tool/Platform	Purpose	Justification
Smart Watch + Health SDK	Sensor data collection (HR, BP, SpO2, ECG)	Industry-standard wearable with medical-grade sensors and WearOS SDK
Python / TensorFlow / scikit-learn	ML model development and training	Industry-standard ML framework with extensive biomedical support
TensorFlow Lite	On-device Android ML inference	Enables low-latency inference without cloud dependency
Android Studio (Kotlin)	Android app development	Official IDE for Wear OS and Android app development
Firebase Realtime Database	Cloud data sync and alert propagation	Low-latency cloud sync; integrates with other Safe Band components
Firebase Cloud Messaging (FCM)	Push alert delivery	Reliable push notification delivery to hospital dashboard
PhysioNet Datasets	ML model training data	Gold-standard annotated clinical ECG/vital sign datasets

Table 2: Tools and Platforms

Validation Strategy and Performance Metrics

The proposed system will be evaluated using appropriate validation techniques, including cross validation and testing on separate datasets. Model performance will be assessed using standard evaluation metrics such as accuracy, precision, recall, and F1-score.

In addition to model level evaluation, system level performance will also be assessed in terms of response time, reliability, and usability. User experience and acceptance will be evaluated through structured feedback mechanisms. The evaluation aims to determine the effectiveness and practicality of the proposed system within a real-world context.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of proposed solution and its components

The proposed SafeBand system is designed as an end-to-end IoT-based health monitoring platform that enables continuous physiological data collection, real-time analysis, and emergency response coordination. The system integrates wearable devices, cloud infrastructure, machine learning components, and user-facing applications into unified architecture.

The ML-Driven Emergency Detection System is the core analytical component of architecture. It is responsible for processing continuous physiological data streams and identifying potential emergency health conditions. Upon detection of a critical event, the system is designed to generate alerts and communicate relevant information to connected components, including mobile applications and emergency response systems.

System Components

Component	Primary Role	Key Output
ML Emergency Detection (This component)	Detect critical patient events from vitals	Alert → App + Dispatch
ML Ambulance Dispatch	Route nearest ambulance to patient GPS	Dispatch instruction
ML Ward Monitoring	Monitor in-hospital ward patients	Ward status dashboard
ML Prediction System	Predict future health deterioration	Risk prediction report

Table 3: System Components

System diagram of components

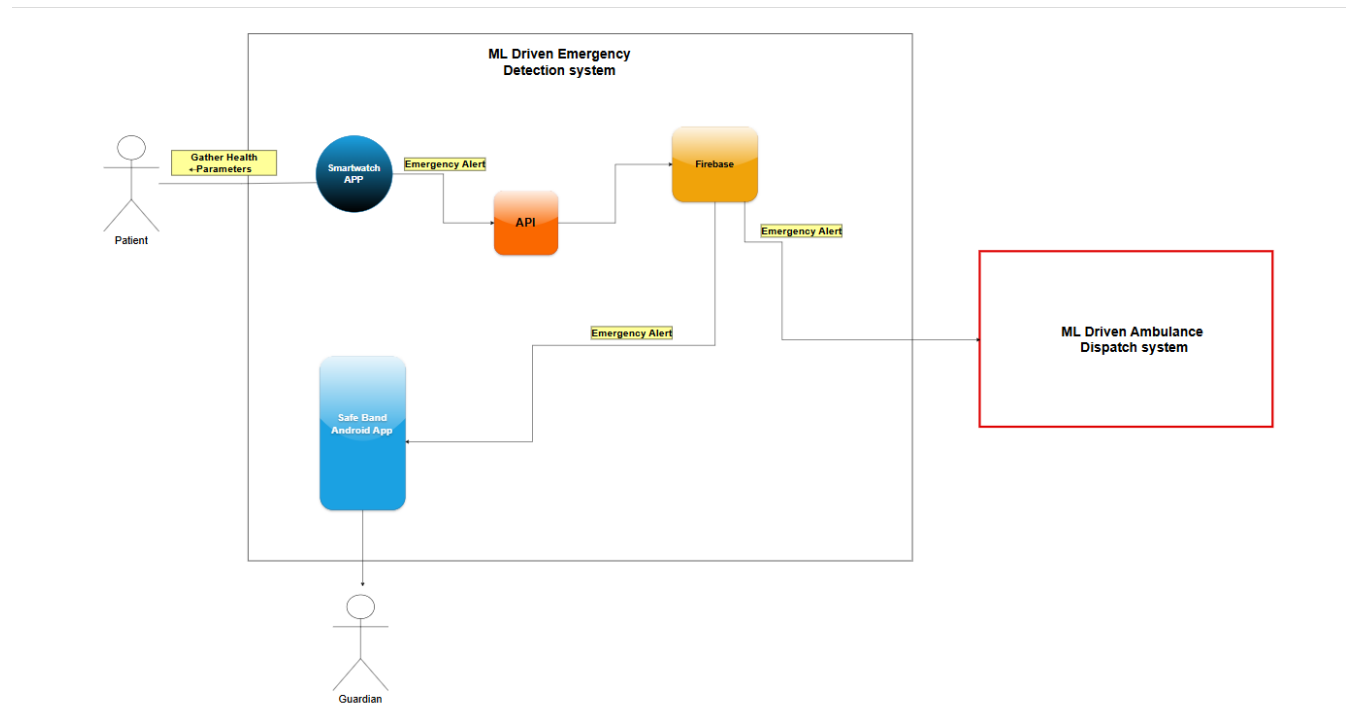


Figure 1: System diagram of components

Highlight the individual component and its parts

Emergency Detection Component

The ML Emergency Detection System is implemented primarily within the mobile application, where real-time physiological data received from the wearable device is processed and analyzed. The component consists of the following functional sub-components:

- **Data Acquisition Module:** Receives real-time physiological data (heart rate, blood pressure, SpO₂, and GPS) directly from the wearable device to the mobile application.
- **Preprocessing Module:** Cleans and prepares incoming sensor data by handling noise, missing values, and formatting it into a suitable structure for analysis.
- **Feature Processing Module:** Extracts meaningful features from the physiological signals to enhance model performance.
- **On-Device Classification Model:** Applies machine learning algorithms within the mobile application to analyze incoming data and identify potential emergency conditions.
- **Decision Module:** Evaluates the model output and determines whether the detected condition represents a potential critical event requiring further action.
- **Alert Transmission Module:** Sends detected emergency alerts from the mobile application to the cloud database (Firebase) for further propagation.

- **External Communication Module:** Forwards alert information via APIs from the cloud layer to external systems such as hospital dashboards and ambulance dispatch services.

Component interactions

The component interactions of the Safe Band system illustrate how real-time monitoring and emergency detection are achieved through coordinated communication between system components.

The wearable device continuously collects physiological data such as heart rate, blood pressure, blood oxygen saturation (SpO₂), and location information. This data is transmitted directly to the mobile application, where the ML-based Emergency Detection System is implemented.

Within the mobile application, the received data is processed and analyzed using machine learning techniques to identify potential abnormal or critical health conditions. When a potential emergency is detected, the application generates an alert containing relevant patient information.

This alert is then transmitted to the cloud database, which acts as an intermediary communication layer. From there, the alert information is forwarded to external systems, including the ambulance dispatch system and hospital monitoring dashboards, via appropriate communication interfaces.

At the same time, the mobile application provides real-time feedback and notifications to the patient or caregiver. This coordinated interaction between the wearable device, mobile application, cloud infrastructure, and external systems enables timely and effective emergency response.

Data flow explanation

The ML Emergency Detection System follows a structured data processing pipeline that begins with physiological data collection and ends with emergency alert generation and communication.

Initially, physiological data such as heart rate, blood pressure, SpO₂, and location are collected by the wearable device and transmitted to the mobile application. The mobile application processes this data through preprocessing and feature extraction stages before applying the machine learning model for classification.

Based on the model output, the system determines whether the patient's condition is normal or potentially critical. If a critical condition is identified, an alert is generated within the application.

The alert is then transmitted to the cloud database, from which it is communicated to external systems such as ambulance dispatch services and hospital dashboards. Simultaneously, notifications are displayed within the mobile application.

This data flow supports continuous monitoring and enables timely communication of critical health events.

Scalability and security considerations

Scalability

The system is designed to support multiple users by handling data streams independently for each patient. Cloud-based infrastructure allows the system to scale as the number of users increases, ensuring that data processing and communication remain efficient.

The architecture supports parallel processing of multiple data streams, enabling real-time monitoring across a large number of users. Additionally, system performance can be further enhanced through periodic model updates based on accumulated data.

Security

The system incorporates standard security practices to ensure the protection of sensitive patient data. Data transmission between system components is secured using encrypted communication protocols.

Access to system resources is controlled through authentication and authorization mechanisms, ensuring that only authorized users can access patient data.

Furthermore, data privacy is maintained through anonymization techniques and secure storage practices. Sensitive information, such as patient location and health data, is shared only with authorized entities when necessary for emergency response.

System activities are logged to maintain traceability and support monitoring and auditing processes.

5. USER REQUIREMENTS

Stakeholder identification

- i. Patients: Individuals with chronic cardiac, hypertensive, or respiratory conditions who wear the smart Watch continuously.
- ii. Attending Physicians / Nurses: Healthcare providers who receive emergency alerts via the hospital dashboard and require accurate, timely patient status information.
- iii. Hospital Administrators: Managers responsible for monitoring system uptime, patient coverage, and alert log review.
- iv. System Operators / IT Staff: Personnel responsible for Firebase configuration, ML model updates, and application maintenance.

Functional requirements

- i. Continuous Heart Rate Monitoring – The system shall continuously collect heart rate readings from the smartwatch.
- ii. SpO2 Monitoring – The system shall monitor blood oxygen levels at regular intervals and on-demand.
- iii. Blood Pressure Collection – The system shall collect blood pressure data at configurable intervals and on-demand.
- iv. ECG Acquisition – The system shall record a 30-second ECG upon detecting abnormal heart rate or manual activation.
- v. Real-Time ML Classification – The system shall classify physiological data into predefined health states within a short time period after data collection.
- vi. Emergency Alert Generation – The system shall generate emergency alerts for abnormal health states within a few seconds of detection.
- vii. Firebase Synchronization – The system shall synchronize patient vitals and alert events to the Firebase database in real time.
- viii. Alert Notification – The system shall notify designated healthcare providers via push notifications in case of emergencies.
- ix. Patient Dashboard Display – The system shall display current vitals, alert status, and history on the Android app for patients or carers.
- x. Offline Operation – The system shall continue monitoring and local alerting during internet outages, syncing data once connectivity is restored.

Non-functional requirements

- i. Performance: ML inference and alert notification shall occur within acceptable time limits to ensure timely response.
- ii. Accuracy: The classification system shall achieve high accuracy and minimize false positives.
- iii. Reliability: The system shall maintain high uptime and ensure minimal data loss.
- iv. Security: All patient data shall be encrypted in transit and at rest; access shall be controlled via authentication.
- v. Usability: The Android app shall provide an intuitive interface suitable for patients and carers.
- vi. Battery Efficiency: Continuous monitoring shall have minimal impact on smartwatch battery life.
- vii. Portability: The application shall support recent Android and Wear OS versions.

6. COMMERCIALIZATION PLAN

The primary target market is private and semi-government hospitals in Sri Lanka seeking to expand remote patient monitoring capabilities. Secondary markets include corporate wellness programs, insurance companies offering health monitoring incentives, and individual consumers with chronic cardiac conditions (estimated 2.3 million individuals in Sri Lanka with hypertension alone, per the Ministry of Health 2022 report).

Value Proposition

Safe Band offers clinically validated, ML-powered emergency detection integrated with hospital workflows a capability unavailable in any existing consumer wearable on the Sri Lankan market. By reducing emergency response time through automated detection and dispatch integration, the system has the potential to significantly improve survival outcomes for cardiac and respiratory emergencies, delivering measurable clinical and financial value to healthcare providers.

Revenue Model

1. B2B SaaS Subscription: Hospitals and clinics pay a monthly per-patient monitoring fee (estimated LKR 1,500–3,000/patient/month) for access to the hospital dashboard, alert management, and monitoring infrastructure.
2. B2C Direct: Individual consumers purchase the Safe Band app subscription (estimated LKR 500/month) for personal emergency monitoring, paired with their existing or purchased Smartwatch.
3. Insurance Partnership: Revenue sharing with health insurers who offer premium discounts to policyholders using Safe Band monitoring.

Competitive Advantage

1. Local calibration: ML models validated against Sri Lankan physiological baselines and clinical contexts.
2. Integrated ecosystem: Unique integration of detection, hospital alerting, and ambulance dispatch in a single platform.
3. Existing hardware leverage: Uses the Smart watch, which many target customers already own or can purchase through retail channels.
4. Clinical explainability: Dashboard provides parameter-level alert rationale rather than black-box alerts.

7. BUDGET AND JUSTIFICATION

Item	Total (LKR)
Smartwatch	60,000
Firebase Blaze Plan (12 months)	15,000
Android Studio (Free / Open Source)	0
TensorFlow / Python Libraries	0
PhysioNet Dataset Access (Open)	0
Travel	6000
TOTAL	LKR 81000.00

Table 6: Budget and justification

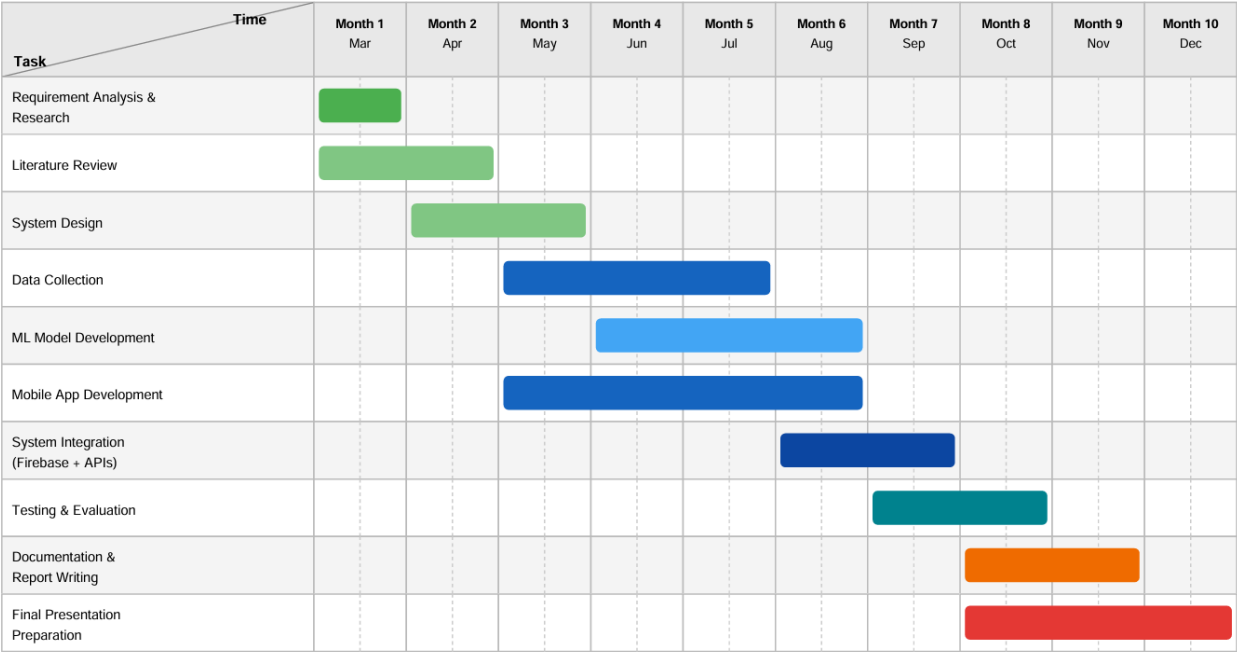
8. WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Task / Sub-Task	Duration	Deliverable
1.0	REQUIREMENT ANALYSIS & RESEARCH	1 Month	
1.1	Define problem statement & project scope	1 week	Problem Statement Doc
1.2	Stakeholder analysis (patients, caregivers)	1 week	Stakeholder Map
1.3	Identify functional & non-functional requirements	1 week	Requirements Spec
1.4	Proposal writing & submission	1 week	Proposal Document
2.0	LITERATURE REVIEW	2 Months	
2.1	Review IoT wearable health monitoring systems	2 weeks	Literature Notes
2.2	Review ML algorithms for health alert prediction	2 weeks	ML Survey Summary
2.3	Study DSR methodology framework	1 week	DSR Framework Notes
2.4	Review Sri Lanka healthcare & emergency context	1 week	Context Report
2.5	Compile related works & identify research gap	1 week	Literature Review Chapter
3.0	SYSTEM DESIGN	2 Months	
3.1	Define system architecture (IoT + Cloud + App)	1 week	Architecture Diagram
3.2	Select hardware components & sensors	1 week	Hardware Spec Sheet
3.3	Design Firebase database schema	1 week	DB Schema Diagram
3.4	UI/UX wireframe design (mobile app screens)	2 weeks	Wireframe Mockups
3.5	API architecture & integration plan	1 week	API Design Document
4.0	DATA COLLECTION	3 Months	
4.1	Set up IoT sensor hardware prototype	2 weeks	Hardware Prototype
4.2	Collect sensor data (heart rate, SpO2, temp)	3 weeks	Raw Sensor Dataset
4.3	Recruit participants & collect user health data	4 weeks	User Data Records
4.4	Data cleaning, labelling & preprocessing	2 weeks	Cleaned Dataset
4.5	Dataset validation & quality check	1 week	Validated Dataset
5.0	ML MODEL DEVELOPMENT	3 Months	
5.1	Feature engineering & selection	2 weeks	Feature Set
5.2	Train ML models (Random Forest, LSTM, etc.)	3 weeks	Trained Models
5.3	Model evaluation (accuracy, precision, recall, F1)	2 weeks	Evaluation Report
5.4	Hyperparameter tuning & model optimization	1 week	Optimized Model

5.5	Alert threshold calibration	1 week	Threshold Config File
6.0	MOBILE APP DEVELOPMENT	4 Months	
6.1	Development environment setup (Flutter/React Native)	1 week	Dev Environment
6.2	Implement core screens (dashboard, profile, history)	3 weeks	Core App Screens
6.3	Real-time data display & health metrics UI	2 weeks	Live Data UI
6.4	Emergency alert & push notification module	2 weeks	Alert Module
6.5	Location & emergency contact management	1 week	Contact Feature
6.6	UI polishing & accessibility improvements	2 weeks	Polished App Build
7.0	SYSTEM INTEGRATION (FIREBASE + APIs)	2 Months	
7.1	Integrate Firebase Auth, Firestore & Realtime DB	2 weeks	Firebase Integration
7.2	Deploy ML model as REST API (Flask/FastAPI)	2 weeks	ML API Endpoint
WBS ID	Task / Sub-Task	Duration	Deliverable
7.3	Connect IoT device ↔ Firebase data pipeline	2 weeks	IoT-Cloud Pipeline
7.4	Integrate ML API with mobile app	1 week	Integrated System
7.5	End-to-end system testing (device → cloud → app)	1 week	Integration Test Report
8.0	TESTING & EVALUATION	2 Months	
8.1	Unit testing (sensor modules, API endpoints)	1 week	Unit Test Results
8.2	Integration & system testing	1 week	System Test Report
8.3	User acceptance testing (UAT) with participants	2 weeks	UAT Feedback Report
8.4	Performance & latency benchmarking	1 week	Benchmark Results
8.5	Bug fixing, refinements & re-testing	1 week	Final Build
9.0	DOCUMENTATION & REPORT WRITING	2 Months	
9.1	Write & finalize all thesis chapters	3 weeks	Draft Thesis Chapters
9.2	Compile final project report	1 week	Final Report
9.3	Write research paper (IEEE/Springer format)	2 weeks	Research Paper Draft
9.4	Build & deploy project website	2 weeks	Live Project Website
9.5	Complete project logbook & checklist	1 week	Logbook & Checklist
10.0	FINAL PRESENTATION PREPARATION	3 Months	
10.1	Prepare final presentation slides	1 week	Presentation Slides
10.2	Prepare live system demo	1 week	Demo System

Table 7: Work Breakdown Structure

9. GANTT CHART



10. REFERENCES LIST

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11.APPENDICES

- Preliminary UI mockups

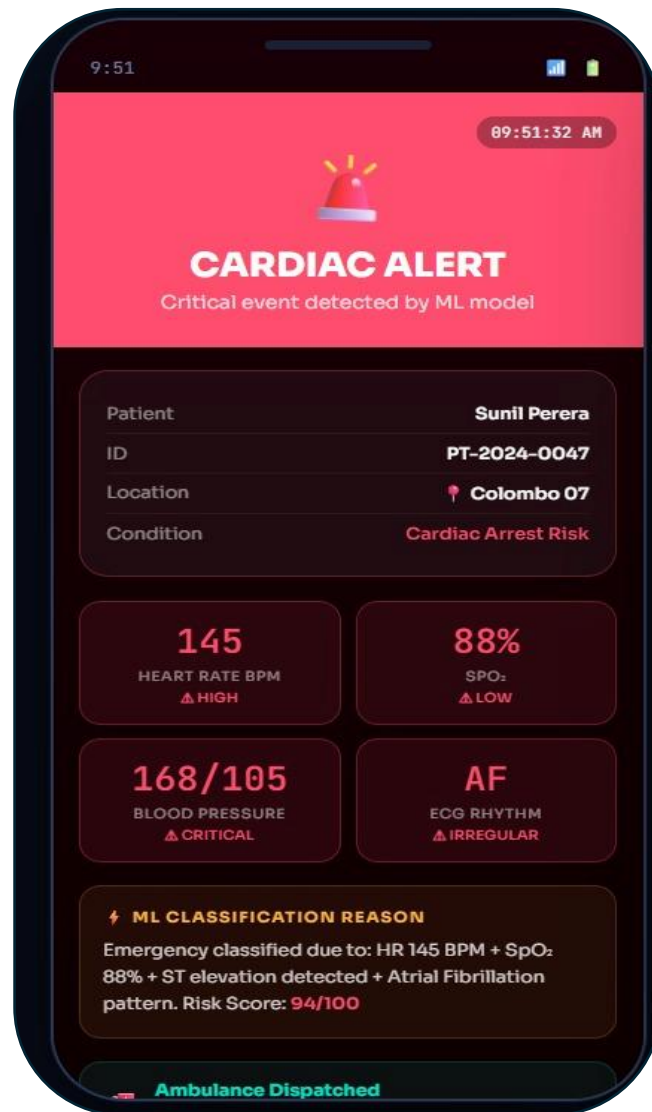


Figure 4: Preliminary UI

